



SRI SRI UNIVERSITY

FACULTY OF ENGINEERING & TECHNOLOGY

MAJOR PROJECT REPORT

Social Media Bot Detection w/ NLP

Baccalaureus Technologiae

Computer Science & Engineering

Specialization in DataScience

Submitted by:

Lokesh Patra | FET-BDS-2022-26-020

Indigibilli Harshit | FET-BDS-2022-26-015

Rudra Narayan Behera | FET-BDS-2022-26-017

Mayank Sharad Kapse | FET-BDS-2022-26-018

Subham Nayak | FET-BDS-2022-26-019

Industry Expert

Dr. Kavitha

Subject Matter Expert

IBM

Guided By

Dr. Saumendra Kumar Mohapatra

Asst. Prof., Faculty of Engineering & Technology

Sri Sri University

Certificate of Submission

This is to certify that **Lokesh Patra, Indigibilli Harshit, Rudra Narayan Behera, Mayank Sharad Kapse, and Subham Nayak**, students of BTech. in Computer Science and Engineering at Sri Sri University, have successfully completed and submitted their major project titled " **Social Media Bot Detection w/ NLP** " as part of their academic curriculum.

The project was officially submitted on **April 15, 2025**.

Throughout the course of this project, the students demonstrated a *commendable grasp* of the subject matter and showcased *exceptional skills* in **research, analysis, and presentation**.

We hereby acknowledge their efforts and dedication toward completing this project and commend their commitment to academic excellence.

Signature
Internal Supervisor

Signature
External Examiner

Acknowledgement

We would like to express our sincere gratitude to all those who contributed to the success of this project. First and foremost, we extend our heartfelt thanks to the management and Experts team for their continuous support and encouragement throughout the project.

We are immensely grateful to our team members for their dedication, hard work, and collaboration, which played a pivotal role in developing and implementing the bot detection model. Their expertise, creativity, and commitment have been instrumental in achieving our project objectives.

We would like to acknowledge the invaluable support and guidance provided by our mentor, **Dr. Saumendra Kumar Mohapatra**, and Subject Matter Expert, **Dr. Kavitha**. Their insights, feedback, and expertise have significantly enriched the project and enhanced its impact.

We extend our appreciation to all stakeholders, including our industry partners, **IBM**, for their valuable inputs, data, and resources that contributed to the success of the project. Their collaboration and cooperation have been essential in understanding industry requirements and implementing effective solutions.

Last but not least, we acknowledge the support of our families, friends, and colleagues who stood by us and provided encouragement and understanding throughout the project journey.

Thank you to everyone involved for their contributions, dedication, and commitment to excellence, making this project a success.

Regards,

Lokesh Patra

Indigibilli Harshit

Rudra Narayan Behera

Mayank Sharad Kapse

Subham Nayak

Abstract

In today's digital era, social media platforms such as Twitter have become a central medium for information sharing, public engagement, and opinion formation. However, these platforms are increasingly plagued by social bots—automated accounts designed to mimic human behaviour to manipulate discussions, spread misinformation, or amplify specific content.

This project aims to develop a comprehensive system for detecting such social media bots using Natural Language Processing (NLP) and ensemble machine learning techniques. Utilizing a web-crawled dataset of over 396,000 user profiles and tweets, the system extracts and engineers numerous behavioural and linguistic features to train models.

A Voting Classifier combining Random Forest and Gradient Boosting classifiers achieved a detection accuracy of over 92% with a ROC AUC score of 0.96. SHAP (SHapley Additive exPlanations) values were employed for model interpretability, identifying the most influential features in classification.

The system is evaluated on various testing criteria including accuracy, performance, robustness, and explainability. The results demonstrate the potential of ensemble learning and feature-rich NLP approaches for real-world social media bot detection.

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Introduction

The proliferation of bots on social media platforms has raised significant concerns regarding the authenticity of online interactions, the manipulation of public discourse, and the spread of disinformation. Bots are automated software agents capable of performing tasks like tweeting, liking, or following with the intent to simulate human-like activity. While some bots serve legitimate purposes, malicious bots are commonly used in large numbers to skew trending topics, impersonate real users, and amplify polarizing content.

This project focuses on developing a scalable and explainable system to detect such bots on Twitter using Natural Language Processing (NLP) and machine learning. The approach involves collecting and processing a large dataset of user accounts and tweets, engineering a diverse set of behavioural and content-based features, and training a hybrid ensemble model to classify accounts as bots or humans. Advanced visualization tools such as ROC curves and SHAP summary plots were used to validate model performance and interpretability. The primary goal is to achieve high detection accuracy while ensuring the system remains interpretable, generalizable, and robust across varied types of accounts and behaviours.